

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Artificial Intelligence and Data Science
ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 1 – Constraint-Based Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	AI&DS-2024	Date:	

Code of Conduct

I, Y. Harsha Vardhan Reddy (192425227) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ____

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.	Q2
Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

Annexure A – Constraint-Based Problem Solving

1. Given the following paragraph:

"John and Mary went to the park. He brought a ball. She wanted to play with it. The dog chased him excitedly. Finally, they all went home."

Tasks: a) Identify all referring expressions (mentions) and possible antecedents.

b) Apply the following **constraints** to resolve the references:

- Gender and number agreement
- Recency (prefer recent antecedents)
- Semantic compatibility (e.g., animacy and action suitability)
- Coherence (maintain consistent entity chains)

c) Draw a **constraint graph** or table showing variables, domains, and which constraints eliminate possible links.

d) Provide the final resolved coreference chains and rewrite the paragraph with clear references.

e) Justify the priority order of your constraints and discuss what would happen if the recency constraint is relaxed.

2. Conversation History: User: "I have an important exam tomorrow but I'm not able to concentrate."

Required Dialog Act: Advise + Encourage (Give advice and motivate the student).

Constraints:

1. Maintain **entity coherence** (reuse "exam", "concentrate", "you").
2. Use **discourse relations** (Cause-Effect or Elaboration).
3. Include at least **two** of these keywords: focus, break, confident.
4. Response length: 2–3 sentences only.
5. Logical consistency (no contradictory statements).
6. Politeness and positive tone.

Tasks: a) Identify the relevant discourse planning steps needed before generation.

b) Generate **three** different possible responses that satisfy **all** the above constraints.

c) Evaluate the three responses and select the best one. Justify your choice using coherence and constraint satisfaction criteria.

d) Explain how violating any two constraints would affect the quality of the dialog.

3. **Source Sentence:** “The bank by the river flooded after the storm, but it was saved by quick action.”

Note: “bank” is ambiguous (financial institution or riverbank).

Constraints:

1. Correct **Word Sense Disambiguation** using context.
2. Create a **predicate logic** representation (e.g., using predicates like flood(x), location(x,y)).
3. Maintain **discourse structure** (Contrast relation between the two clauses).
4. Preserve all important entities and relations (no information loss).
5. Ensure **text coherence** in the generated output.

Tasks: a) Resolve the ambiguous word “bank” and justify using constraints.

b) Write a predicate logic representation of the sentence.

c) Generate a good **target language sentence** (in any Indian language or simple English paraphrase) that satisfies all constraints.

d) Draw a simple **discourse tree** (RST-style) showing the rhetorical relation.

e) Discuss the advantages of using constraint-based approach over pure statistical machine translation for this sentence

1. Coreference Resolution Using Constraints

Given paragraph: “John and Mary went to the park. He brought a ball. She wanted to play with it. The dog chased him excitedly. Finally, they all went home.”

1(a) Identifying referring expressions and possible antecedents

A referring expression is a word or phrase whose interpretation depends on another entity in the discourse. Pronouns such as he, she, it, him, and they are the most obvious referring expressions in this paragraph. The noun phrase “the dog” also introduces a new discourse entity, while “they all” refers back to a group of previously introduced entities.

Mention	Type	Possible antecedents	Best candidate	Reason
John	Proper noun	—	John	Introduces male singular entity
Mary	Proper noun	—	Mary	Introduces female singular entity
He	Pronoun	John, Mary	John	Male singular agreement
a ball	Noun phrase	—	ball	Introduces inanimate object
She	Pronoun	John, Mary	Mary	Female singular agreement

it	Pronoun	ball, Mary, John	ball	Neuter/inanimate object and play compatibility
the dog	Noun phrase	—	dog	Introduces animal entity
him	Pronoun	John, dog	John	Male singular; dog has no stated male gender
they	Plural pronoun	John + Mary; John + Mary + dog	John + Mary + dog	Plural group and final discourse coherence
all	Group modifier	John, Mary, dog	John + Mary + dog	Signals the whole salient group

1(b) Applying the constraints

1. Gender and number agreement

Gender and number agreement is the first strong filtering constraint. “He” must normally refer to a singular male entity. John is male by the name and discourse convention, whereas Mary is female. Therefore, Mary is eliminated as an antecedent for “he.” Similarly, “she” must refer to a singular female entity, making Mary the natural antecedent and eliminating John. For “him,” John is the strongest candidate because he is a singular male human and the paragraph does not establish that the dog is male. The plural pronoun “they” cannot refer to only John, Mary, or the dog individually; it requires a plural discourse group.

2. Recency

Recency prefers an antecedent that was mentioned recently, provided that other constraints do not contradict it. For example, when “it” occurs, the immediately preceding salient object is “a ball,” so the ball is preferred. Recency is useful because writers commonly continue discussing the most recently active entity. However, recency should not override grammatical agreement or semantic compatibility. A recent candidate that violates gender, number, or meaning should be rejected.

3. Semantic compatibility

Semantic compatibility checks whether the candidate makes sense with the predicate or action. The phrase “play with it” naturally accepts an object such as a ball, while John or Mary would be much less appropriate as the intended referent. Likewise, “the dog chased him” strongly favors a human male referent for “him” because the text has already established John and Mary as people. The dog itself is the grammatical subject of “chased,” so interpreting “him” as the dog would be semantically and structurally awkward.

4. Coherence

Coherence requires the resolved links to create stable entity chains across the paragraph. The paragraph first introduces John and Mary, then gives actions involving them, introduces a ball, and later introduces a dog. A coherent interpretation is that John brought the ball, Mary wanted to play with the ball, the dog chased John, and finally John, Mary, and the dog went home together. This produces a consistent discourse model without unexplained shifts in reference.

1(c) Constraint graph / variable-domain table

The following table represents the problem as a constraint satisfaction task. Each referring expression is treated as a variable, its possible antecedents form its domain, and the constraints successively eliminate incompatible links.

Variable	Initial domain	Gender/number	Recency	Semantic/coherence	Final value
He	{John, Mary}	Mary eliminated	John remains salient	Person/agent fits	John
She	{John, Mary}	John eliminated	Mary salient	Person/agent fits	Mary
it	{John, Mary, ball}	Ball fits neuter	Ball is most recent object	Play-with-object fits	ball
him	{John, dog}	John preferred; dog gender unstated	Dog is recent but is subject	Dog-chased-person fits	John
they	{John+Mary, John+Mary+dog}	Plural required	Recent dog + existing pair	Final “all” favors group	John+Mary+dog

A compact graph interpretation is: $\text{He} \rightarrow \text{John}$; $\text{She} \rightarrow \text{Mary}$; $\text{it} \rightarrow \text{ball}$; $\text{him} \rightarrow \text{John}$; $\text{they} \rightarrow \{\text{John, Mary, dog}\}$. Each arrow is retained only when the relevant agreement, semantic, recency, and coherence constraints are satisfied.

1(d) Final coreference chains

John-chain: $\text{John} \rightarrow \text{He} \rightarrow \text{him} \rightarrow \text{they}$

Mary-chain: Mary → She → they

Ball-chain: a ball → it

Dog-chain: the dog → they

Rewritten paragraph with clear references

John and Mary went to the park. John brought a ball. Mary wanted to play with the ball. The dog chased John excitedly. Finally, John, Mary, and the dog all went home.

1(e) Priority order of constraints and effect of relaxing recency

A practical priority order is: (1) hard grammatical agreement, (2) semantic compatibility, (3) coherence, and (4) recency as a preference. Gender and number agreement should normally have the highest priority because a candidate that violates agreement is grammatically inappropriate. Semantic compatibility comes next because a grammatically possible antecedent may still be semantically impossible or highly unlikely. Coherence follows because the selected links should form a stable discourse model. Recency is valuable but should be treated as a soft preference rather than an absolute rule.

If recency is relaxed, the system becomes more willing to choose an older antecedent when that older entity is semantically or discourse-wise better. In this paragraph, relaxing recency would not change the strongest resolutions of “He,” “She,” or “it,” because agreement and semantics are already decisive. For “him,” relaxing recency is actually useful: although “the dog” is the nearest noun phrase, the system can select John because John is the compatible male human entity and the dog is already the subject of “chased.” For “they,” relaxing strict recency allows the system to recover the larger discourse group John + Mary + dog instead of selecting an overly narrow recent antecedent.

2. Constraint-Based Dialog Act Planning

Conversation history: User: “I have an important exam tomorrow but I’m not able to concentrate.”

Required dialog act: Advise + Encourage.

2(a) Relevant discourse planning steps

Identify the user's main problem: an important exam is tomorrow and the user cannot concentrate.

Identify the communicative goal: provide practical advice while also motivating and encouraging the student.

Maintain entity coherence by continuing to refer to the same entities and concepts: “you,” “exam,” and “concentrate.”

Select a discourse relation. A Cause-Effect structure is appropriate: lack of concentration can be addressed by taking a short break and reducing distractions, which can improve focus.

Select the required lexical constraints. At least two of the keywords focus, break, and confident must appear.

Construct a logically consistent response in which the advice and encouragement support each other.

Check politeness and positive tone so that the response motivates rather than criticizes the student.

Check length: exactly 2–3 sentences.

Perform a final constraint check before generation.

2(b) Three possible responses satisfying all constraints

Response 1

Take a short break, remove distractions, and return to your exam preparation with one small topic at a time so you can focus better. Remember that you have prepared for this exam, so stay confident and keep going.

Response 2

Because the exam is tomorrow, take a short break and then study in small focused sessions to improve your concentration and focus. You can handle the exam, so stay confident and do your best.

Response 3

If you are unable to concentrate, take a short break, drink some water, and start again with one easy exam topic to regain focus. Trust your preparation and stay confident; steady effort now will help you feel more ready for the exam.

2(c) Evaluation and selection of the best response

Response	2–3 sentences	Advice	Encourage	Keywords	Coherence	Overall
1	Yes	Yes	Yes	break, focus, confident	Strong	Very good
2	Yes	Yes	Yes	break, focus, confident	Strong	Very good
3	Yes	Yes	Yes	break, focus, confident	Very strong	Best

Best response: Response 3. It satisfies every explicit constraint while providing a clear cause-and-effect progression: difficulty concentrating leads to a short break, followed by a manageable study action, which helps restore focus. It maintains the same entities and topic throughout, uses all three permitted keywords, and ends with encouragement without making an unrealistic promise. The wording is also polite and supportive.

2(d) Effect of violating two constraints

Violation 1: Entity coherence

If entity coherence is violated, the response may suddenly shift away from the user's exam or concentration problem. For example, a response that starts discussing unrelated travel or sports would not connect properly to the conversation. The user would have to infer why the new topic was introduced, reducing relevance and making the dialog feel disconnected.

Violation 2: Response length

If the 2–3 sentence length constraint is violated, the response may become too long and lose its role as a quick supportive answer. A very long explanation could overwhelm a student who is already struggling to focus. Conversely, a response that is too short may fail to provide both advice and encouragement. Constraint-based generation therefore controls both completeness and usability.

3. Constraint-Based Word Sense Disambiguation and Discourse Representation

Source sentence: “The bank by the river flooded after the storm, but it was saved by quick action.”

The word “bank” is ambiguous in isolation: it can denote a financial institution or the land beside a river.

The objective is to determine the intended sense while preserving the sentence's logical and discourse structure.

3(a) Resolving the ambiguous word “bank”

The correct sense is RIVERBANK, meaning the land alongside a river. The strongest evidence is the phrase “by the river.” This is a direct collocational and semantic cue. The predicate “flooded” is also highly compatible with a riverbank because riverbanks can be inundated after storms. The expression “saved by quick action” can naturally describe emergency action that protects a riverbank from further damage.

Constraint	Financial bank	Riverbank	Decision	Explanation
Context	Weak	Strong	Riverbank	“by the river” directly selects river-related meaning
Flood compatibility	Possible_but unlikely	Very strong	Riverbank	Riverbanks commonly flood
Storm relation	Weak	Strong	Riverbank	Storms can cause river flooding
Saved by quick action	Possible	Natural	Riverbank	Emergency action can protect a bank/embankment

Overall coherence	Low	High	Riverbank	All parts form one coherent event
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Therefore, the disambiguation chain is: bank $\rightarrow$ riverbank. The financial-institution sense is eliminated because it does not fit the local context as well as the riverbank sense. The constraint-based approach does not rely on the word alone; it considers surrounding entities, predicates, relations, and discourse.

3(b) Predicate logic representation

Let b represent the riverbank, r the river, s the storm, and a the quick action. A suitable representation is:
 $\text{RiverBank}(b) \wedge \text{River}(r) \wedge \text{Storm}(s) \wedge \text{Action}(a) \wedge \text{By}(b,r) \wedge \text{Flooded}(b) \wedge \text{After}(\text{Flooded}(b),s) \wedge \text{Quick}(a) \wedge \text{Saved}(b,a)$

The predicate meanings are: $\text{RiverBank}(b)$ states that b is a riverbank; $\text{By}(b,r)$ states that the bank is located beside the river; $\text{Flooded}(b)$ states that the bank flooded; $\text{After}(\text{Flooded}(b), s)$ connects the flooding event to the storm; $\text{Quick}(a)$ states that the action was quick; and $\text{Saved}(b,a)$ states that the bank was saved through that action.

Discourse-level predicate

$\text{Contrast}(\text{FloodingEvent}, \text{SavingEvent})$

This captures the function of “but.” The first clause describes a damaging event, while the second clause introduces a contrasting positive outcome: despite the flooding, quick action saved the bank.

3(c) Target-language sentence / simple English paraphrase

Simple English paraphrase: “The riverbank near the river flooded after the storm, but quick action saved it.”

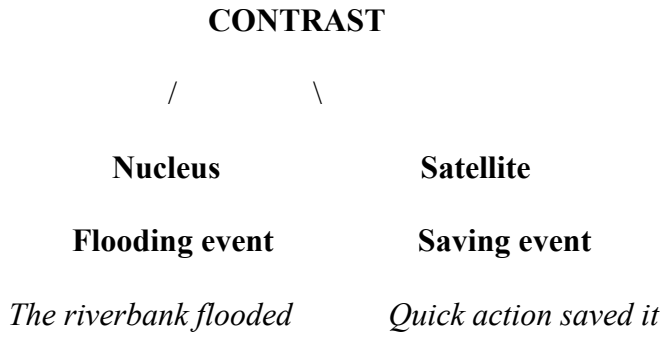
This paraphrase explicitly uses “riverbank” instead of the ambiguous “bank,” preserves the river, storm, flooding, quick action, and saving relationship, and keeps the contrast signaled by “but.” The pronoun “it” clearly refers to the riverbank, so no important entity is lost.

Example in Telugu: “నది పక్కన ఉన్న గట్టు తుఫాను తర్వాత వరదకు గురైంది, కానీ త్వరిత చర్య వల్ల అది రక్షించబడింది.”

The Telugu version preserves the same essential meaning: the riverbank flooded after the storm, but it was saved because of quick action.

3(d) Simple RST-style discourse tree

The sentence contains two main clauses connected by the discourse relation CONTRAST:



In this simplified RST representation, the two clauses are linked by CONTRAST. The first clause introduces the problem or adverse event, while the second clause gives a contrasting outcome in which the bank is saved. The conjunction “but” is the explicit linguistic signal for this relation.

3(e) Advantages of constraint-based approach over pure statistical machine translation

1. Better ambiguity resolution

A constraint-based system explicitly checks contextual clues such as “by the river” and “flooded.” It can therefore choose riverbank rather than financial institution. A purely statistical system may select a frequent translation of “bank” if context modeling is weak or training evidence is insufficient.

2. Preservation of logical meaning

Predicate and semantic constraints make the system explicitly preserve relations such as bank–river, flood–storm, and action–saving. This reduces the chance that an important relationship disappears during translation.

3. Strong discourse coherence

The word “but” establishes a contrast between flooding and saving. A constraint-based approach can represent this relation explicitly and ensure that the target sentence retains the same rhetorical structure.

4. Entity consistency

Once “bank” is resolved as riverbank, later references such as “it” must refer to the same entity. Entity-chain constraints help prevent inconsistent translations where “bank” changes meaning midway.

5. Explainability

The system can explain why it selected riverbank: the phrase “by the river,” the flooding predicate, and the storm context all support that interpretation. This is valuable in academic, legal, medical, and other high-precision language applications.

6. Controlled generation

Constraint checking can prevent information loss, contradictions, and invalid structures. The generated sentence must satisfy predefined requirements rather than merely being statistically probable.

7. Better handling of rare contexts

If a sentence contains an unusual combination of words, a constraint system can still reason from explicit semantic rules. It does not have to depend entirely on seeing the exact sentence pattern during training.

8. Easier error diagnosis

When an output is wrong, the violated constraint can be inspected. For example, an incorrect financial-bank translation indicates failure of contextual or semantic compatibility. This makes debugging more systematic.

Pure statistical or neural machine translation is powerful because it learns complex language patterns from large datasets and can produce fluent translations. However, fluency alone does not guarantee correct word sense, logical consistency, or discourse preservation. A constraint-based approach is especially

useful when correctness and explainability are more important than simply producing a statistically likely sentence. In practice, hybrid systems can combine statistical/neural models with explicit constraints to obtain both fluency and controlled semantic accuracy.

Overall Conclusion

Constraint-based problem solving provides a structured way to handle difficult Natural Language Processing tasks. In coreference resolution, constraints such as agreement, semantic compatibility, recency, and coherence help identify the correct antecedent. In dialog generation, constraints ensure that the response satisfies the required dialog act, maintains the conversation topic, uses required vocabulary, and remains concise and polite. For word sense disambiguation, explicit context and semantic constraints correctly identify “bank” as riverbank and preserve the sentence's logical and discourse relations. Overall, constraints make NLP decisions more transparent, consistent, and controllable.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solution Development	Application of Concepts	Justification	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____